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Numerical Methods

Unit 4: Numerical solution of ordinary differential equations (ODEs)

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Outline

Euler's Method

Backward Euler's method

Introduction to Euler's Method

When we cannot solve an ODE analytically, we use numerical approximation methods to obtain an accurate approximation to the solution of an initial value problem (IVP).
The most basic numerical procedure for solving an (IVP) is the **Euler Method**.
Given the IVP:

$$y' = f(t, y), \quad y(t_0) = y_0,$$

the iterative formula for the Euler's method is given by

$$y_{n+1} = y_n + h \cdot f(t_n, y_n),$$

where h is the step size (smaller h usually means better accuracy) and $f(t_n, y_n)$ is the slope at t_n.

Example

Use the Euler method and step sizes $h = 0.05, 0.025, 0.01,$ and 0.001, to find approximate values of the solution of the IVP

$$y' = \frac{t}{y}, \quad y(1) = 2.$$

on the interval $[1, 2]$.

Exact Solution (for comparison)

Separate variables:

$$y\,dy = t\,dt$$

Integrate:

$$\frac{1}{2}y^2 = \frac{1}{2}t^2 + C$$

$$y^2 = t^2 + C$$

Using $y(1) = 2$:

$$4 = 1 + C \Rightarrow C = 3$$

Exact solution:

$$y(t) = \sqrt{t^2 + 3}$$

$$y(2) = \sqrt{7} \approx 2.64575$$

Euler's Method

Euler update formula:

$$y_{n+1} = y_n + h\frac{t_n}{y_n}, \qquad t_{n+1} = t_n + h$$

Initial values:

$$t_0 = 1, \qquad y_0 = 2$$

Number of steps:

$$N = \frac{2-1}{h}$$

Results at $t = 2$

h	Euler Approximation	Error
0.05	2.64303	0.005528
0.025	2.64439	0.002746
0.01	2.64521	0.001094
0.001	2.64570	0.000109

Exact value:

$$\sqrt{7} \approx 2.64575$$

We need **1000 iterations** to reach $t = 2$.
As $h \to 0$, the approximation converges to the exact solution.

Conclusion

- Euler's method gives first-order accuracy.
- The global error is $O(h)$.
- Smaller step sizes significantly improve accuracy.

Backward Euler Approximation

Using the backward difference to approximate the derivative in the differential equation, we get

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1}), \qquad h = t_{n+1} - t_n.$$

- The formula does **not** give an explicit expression for y_{n+1}.
- Instead, it defines y_{n+1} **implicitly** (Implicit Euler Method).
- We must solve an equation to obtain y_{n+1}, which depends entirely on the nature of the function f.

Remarks

The difficulty of computing y_{n+1} depends on the function $f(t, y)$.

- If f is **linear** in y, the equation can often be solved easily.
- If f is **nonlinear**, numerical techniques such as
  - Fixed-point iteration
  - Newton's method

  may be required.

Although is also first-order accurate and requires solving an implicit equation at each step, it plays a fundamental role in numerical analysis because:

- it provides excellent stability properties, particularly for stiff equations
- it forms the foundation of more advanced implicit methods, such as higher-order backward differentiation formulas.

Example

Approximate the solution of the initial value problem, using the **Backward Euler method**

$$y' = \frac{t}{y}, \qquad y(1) = 2$$

on the interval $[1, 2]$.

Apply the Method

Here

$$f(t, y) = \frac{t}{y}.$$

Substituting into the backward Euler formula gives

$$y_{n+1} = y_n + h\frac{t_{n+1}}{y_{n+1}}.$$

This is an **implicit equation** for y_{n+1}.

Solving for y_{n+1}

Multiply both sides by y_{n+1}:

$$y_{n+1}^2 = y_n y_{n+1} + h t_{n+1}.$$

Rearranging,

$$y_{n+1}^2 - y_n y_{n+1} - h t_{n+1} = 0.$$

This is a quadratic equation in y_{n+1}.

Quadratic Solution

Solving the quadratic equation gives

$$y_{n+1} = \frac{y_n \pm \sqrt{y_n^2 + 4ht_{n+1}}}{2}.$$

Since $y > 0$, we take the positive root:

$$\boxed{y_{n+1} = \frac{y_n + \sqrt{y_n^2 + 4ht_{n+1}}}{2}}$$

Algorithm

Choose a step size

$$h = \frac{2-1}{N}.$$

Initialization

$$t_0 = 1, \qquad y_0 = 2$$

Iteration

For $n = 0, 1, \ldots, N-1$

$$t_{n+1} = t_n + h$$

$$y_{n+1} = \frac{y_n + \sqrt{y_n^2 + 4ht_{n+1}}}{2}.$$

Exercise

Use the backward Euler formula and step sizes $h = 0.05, 0.025, 0.01,$ and 0.001 to find approximate values of the solution of the following initial value problem the interval $0 \leq t \leq 2$.

$$y' = 1 - t + 4y, \qquad y(0) = 1.$$

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